

CODEALPHA VIRTUAL INTERNSHIP

ConnectHub Social Media Platform

Full Stack Web Development Internship Presentation

Presenter: Kartikey Gupta | **Roll No:** 25SCS1003004592 | **Section:** 2CSE37
Student ID: CA/DF1/194583 | **Duration:** 10 July 2026 – 10 August 2026

INTERNSHIP & ORGANIZATION PROFILE

CodeAlpha Virtual Internship Overview

Candidate Profile

- **Name:** Kartikey Gupta
- **Roll Number:** 25SCS1003004592
- **Batch & Section:** B.Tech CSE (2025–2029) | 2CSE37
- **Track:** Full Stack Web Development
- **Student ID:** CA/DF1/194583

Host Organization

- **Company:** CodeAlpha Software Development
- **Recognition:** Registered under MCA, Govt. of India
- **Leadership:** Swati Sri (Founder & CEO)
- **Tenure:** 10th July 2026 – 10th August 2026
- **Outcome:** Completion Certificate & Recommendation Letter

PROJECT OVERVIEW: CONNECTHUB

Modern Full-Stack Social Networking Web Application

Core Problem Statement

Modern social web platforms demand seamless state persistence, secure multi-user authentication, low-latency social feed aggregation, and scalable NoSQL database modeling. ConnectHub addresses these requirements in a lightweight, robust full-stack architecture.

Key Objectives

- Implement secure user authentication (JWT in httpOnly cookies) & password encryption.
- Develop post CRUD functionality (Create, Read, Like/Unlike, Comment, Delete).
- Build personalized newsfeed displaying posts from followed users and self.
- Engineered relational social structures (Follow / Unfollow user graph).

TECHNOLOGY STACK & TOOLS

End-to-End Modern Web Stack

Frontend Technologies

- **HTML5 / CSS3:** Modern semantic layout & modern UI design
- **JavaScript (ES6+):** Async/Await, Fetch API, DOM manipulation
- **Responsive Design:** Mobile & desktop compatible views

Backend & Database

- **Node.js:** Asynchronous server runtime
- **Express.js:** Scalable RESTful API route framework
- **MongoDB:** NoSQL document storage engine
- **Mongoose ODM:** Schema validation & ref population

Security & Auth

- **JWT:** JSON Web Tokens stored in HTTP-Only cookies
- **bcryptjs:** Salted password encryption algorithm

Developer Tools

- **Git & GitHub:** Version control repository management
- **Postman:** REST API contract testing & debugging

SYSTEM ARCHITECTURE & KEY FEATURES

3-Tier Modular Structure

Core Platform Features

- **Auth System:** Registration, Login, Protected routes
- **Post Management:** Text & media posts with deletion rights
- **Interactive Feed:** Chronological aggregated user feed
- **Engagements:** Dynamic Like/Unlike & Comment threads
- **Follow Mechanics:** Graph-based Follower/Following links
- **User Profiles:** Customizable profile & activity summaries

API Endpoint Structure

- `/api/auth/*` — Register, Login, Logout, Profile verification
- `/api/posts/*` — CRUD posts, feed retrieval, like & comment handlers
- `/api/users/*` — Follow/Unfollow endpoints, user profiles

KEY LEARNINGS & CONCLUSION

Summary of Internship Achievements

Technical Growth

- Hands-on expertise in MERN/Node-Mongo stack development.
- Mastered cookie-based secure JWT authorization flows.
- Gained deep knowledge in NoSQL schema design & query optimization.

Conclusion

- Successfully delivered full-stack ConnectHub platform within schedule.
- Received official Certificate of Completion and Letter of Recommendation.
- Repository & full source code hosted on GitHub.

Thank You!

Questions & Feedback | Kartikey Gupta (CA/DF1/194583)